

Custom theming with JSON

See Microsoft's full JSON theme schema: [powerbi-desktop-samples/Report Theme JSON Schema at main · microsoft/powerbi-desktop-samples · GitHub](https://github.com/microsoft/powerbi-desktop-samples/blob/main/Report%20Theme%20JSON/Schema%20at%20main%20-%20microsoft/powerbi-desktop-samples/GitHub)

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Navigating to *View > Themes* gives theming options similar to other Microsoft products, where design changes can be applied to a whole Power BI report in one click.

However, Power BI also allows for custom themes to be created and applied with the use of a JSON file, allowing for much deeper theming options.

Why custom themes are useful

Scottish Government has an established Design System, see: [Colour - Design System](#). This means that websites and web applications all have a consistent style and feel.

Implementing a custom JSON theme in Power BI will ensure compatibility with the Design System. The file has to be created just once and can be applied to all Power BI dashboards to be compatible.

Custom themes allow for much deeper customisability than Power BI's theme GUI. This includes support for more colours, customisation of groups of elements (e.g. colour classes, see below), and definition of the style of specific visuals or specific elements of visuals.

Syntax

JSON comes in key-value pairs and may look something like:

```
{  
  "key": "value",
```

"key2": "values"

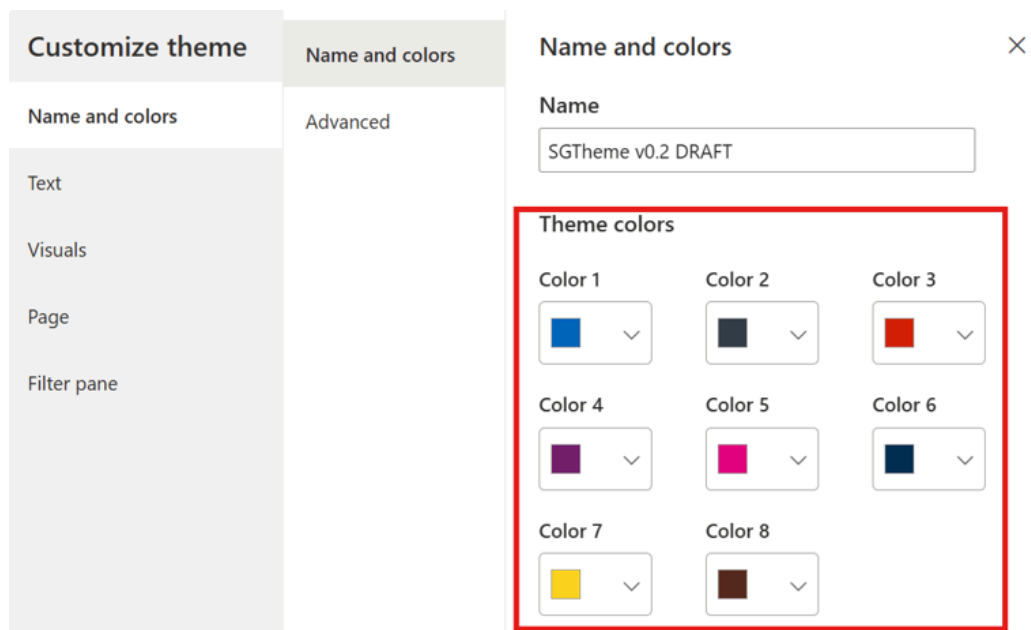
}

Specific key-value pairs work with Power BI custom JSON themes, for details see: [Use report themes in Power BI - Power BI](#)

The only mandatory part of the theme file is "name". E.g. "name": "SG Theme", meaning all other elements are optional for a valid custom JSON theme.

The wildcard "*" can be applied to affect all visual aspects of the dashboard i.e. they are global effects.

"dataColors": Represents colours of shapes used within visuals. These are the 8 colour options which you can choose in the 'Customize theme' menu as seen below:



Note: The JSON file supports technically unlimited amounts of colours

The custom SG Power BI theme

Current state: [SG Theme v1.1.json](#)

Main theme colours

"dataColors": 11 colours have been used from the Design System's data visualisation colour palettes, see: [Data visualisation: colour palettes - Design System](#)

Theme	Colour description	Design System colour reference	Hex code

colour			
1	Categorical data colour 1 (dark blue)	\$ds_datavis__categorical-1	"#002d54"
2	Categorical data colour 2 (teal)	\$ds_datavis__categorical-2	"#2b9c93"
3	Categorical data colour 3 (purple)	\$ds_datavis__categorical-3	"#6a2063"
4	Categorical data colour 4 (orange)	\$ds_datavis__categorical-4	"#e5682a"
5	Categorical data colour 5 (dark green)	\$ds_datavis__categorical-5	"#0b4c0b"
6	Categorical data colour 6 (green)	\$ds_datavis__categorical-6	"#5d9f3c"
7	Categorical data colour 7 (brown)	\$ds_datavis__categorical-7	"#592c20"
8	Categorical data colour 8 (pink)	\$ds_datavis__categorical-8	"#ca72a2"
9	Sequential data colour 2 (blue)	\$ds_datavis__sequential-2	"#0065bd"
10	Sequential data colour 3 (light blue)	\$ds_datavis__sequential-3	"#55a8f2"
11	Non-focussed data colour (grey)	\$ds_datavis__focus--dimmed	"#949494"

Sentiment colours

These set the status colours used by the waterfall chart and the KPI visual.

"good" : "#1a7032" (\$ds_colour__state--positive)

"neutral" : "#000000" (\$ds_colour__black)

"bad" : "#d32205" (\$ds_colour__state--negative)

Gradient colours

These colours get set when choosing gradient conditional formatting.

The sequential data colour palette is used, as usually when applying gradients, the order of data will matter.

"maximum": "#002d54" (\$ds_datavis__sequential-1)

"center": "#0065bd" (\$ds_datavis__sequential-2)

"minimum": "#55a8f2" (\$ds_datavis__sequential-3)

Colour classes

Specific elements of visualisations are categorised as follows:

- firstLevelElements a.k.a. "foreground"
- secondLevelElements a.k.a. "foregroundNeutralSecondary"
- thirdLevelElements a.k.a. "backgroundLight"
- fourthLevelElements a.k.a. "foregroundNeutralTertiary"
- "background"
- secondaryBackground a.k.a. "backgroundNeutral"
- "tableAccent"

i For more details on which visual elements these specifically affect, see: [Use report themes in Power BI Desktop - Power BI | Microsoft Learn](#)

See below customisation of the colour classes, some example effects of the classes are included in the appropriate column:

Colour class	Example effects	Colour used
"foreground"	Text box default colour, data bars axis, slicer dropdown item font colour	"#1a1a1a" (\$ds_colour__grey--dark)
"foregroundNeutralSecondary"	Axis, legend and other labels, slicer outline and slider, button text, icon line and outline	"#5e5e5e" (\$ds_colour__grey)
"backgroundLight"	Axis gridline, table and matrix grid, shape fill	"#ebebcb" (\$ds_colour__background--tertiary)

"foregroundNeutralTertiary"	Disabled button text font and icon line, card category labels, legend dimmed	"#b3b3b3" (\$ds_colour__grey--medium)
"background"	Slicer dropdown items background, button fill, label background	"#ffffff" (\$ds_colour__background--primary)
"backgroundNeutral"	Table and matrix grid outline, disabled button fill and outline colour (when background isn't #ffffff)	"#f8f8f8" (\$ds_colour__background--secondary)
"tableAccent"	The grid outline for table and matrix visuals	"#b3b3b3" (\$ds_colour__border)

Text

See: [Use report themes in Power BI Desktop - Power BI | Microsoft Learn](#)

The following settings have been used for text wherever possible unless otherwise specified:

- "color" : "#1a1a1a" (\$ds_colour__text)
- "fontFace" : "Roboto"

Roboto font is free and open source. It can be downloaded from: [Roboto - Google Fonts](#)

Font sizes used are based on the Design System typography heading sizes.

✕ Click here to view Design System typography specifications

Element	Font size	Weight
H1	30px	bold/700
H2	22px	bold/700
H3	19px	bold/700
H4	16px	bold/700
H5	14px	bold/700
Lead paragraph	19px	light/300

Body	16px	regular/400
Small	14px	regular/400

Note: Certain elements e.g. "largeTitle" may deviate slightly from the Design System by reducing font weight or changing colour. Decisions were made based on User Centred Design Team feedback to improve readability.

The text classes have been implemented as follows:

Text class	Example effects	Attributes
"callout"	Card data labels, KPI indicators	"fontSize": 16 "fontWeight": "400" (Body)
"header"	Key influencers visual title	"fontSize": 19 "fontWeight": "700" (H3)
"title"	Axis titles, multi-row card titles, slicer headers	"fontSize": 16 "fontWeight": "400" (Body)
"largeTitle"	Title of visuals	"fontSize": 19 "fontWeight": "400" (based on H3, font weight reduced)
"label"	Table or matrix row and column headers and values. Text box content, button text	"fontSize": 16 "fontWeight": "400" (Body)
"largeLabel"	Multi-row card visual data labels	"fontSize": 16 "fontWeight": "400" (Body)

"smallLabel"	Slicer date range labels, search box and numeric input. Key influencers visual text content	"fontSize": 14 "fontWeight": "400" (Small)
"lightLabel"	Text box subtitle, slicer dropdown selection, legend text	"fontSize": 16 "fontWeight": "400" (Body)
"boldLabel"	Table and matrix totals/subtotals	"fontSize": 16 "fontWeight": "700" (H4)
"largeLightLabel"	Card and multi-row card category labels, gauge labels	"fontSize": 14 "fontWeight": "400" (Small) "color": "#5e5e5e" (\$ds_colour__grey), text is intended to be secondary and not draw immediate attention
"smallLightLabel"	Data labels, value axis labels	"fontSize": 14 "fontWeight": "400" (Small) "color": "#000000" (\$ds_colour__black), to draw attention, improve readability and reduce ambiguity or misinterpretation of data
"semiboldLabel"	Key influencers visual profile text	"fontSize": 16 "fontWeight": "400" (Body)

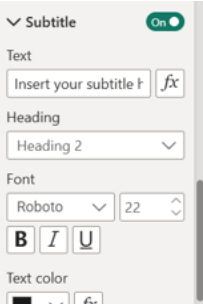
Visual property details

More granular control of specific Power BI elements can be achieved under the "visualStyles" key. It allows for in depth customisation of every type of Power BI chart and other visual elements like text boxes or the filter pane.

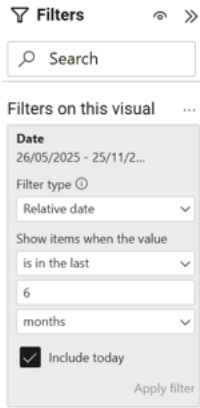
The following settings have been used for text wherever possible unless otherwise specified:

- "color": "#1a1a1a" (\$ds_colour__text)
- "fontFace": "Roboto"

Note: Certain elements e.g. "title" may deviate slightly from the Design System by reducing font weight. This decision was made based on User Centred Design Team feedback. Since a lot of Heading type elements are used in Power BI reports, this was done to improve readability by not overwhelming users with bold text everywhere.

visualStyles class	Affects	Attributes
"title"	Any time a Title is triggered "On" on any visual and text box. 	"show": false (title is hidden by default) "text": "Insert your title here" "heading": "Normal" (Power BI strangely calls H1 'Normal') "fontSize": 30 "bold": false (based on H1, font weight reduced i.e. bold disabled)
"subTitle"	Any time a Subtitle is triggered "On" on any visual and text box. 	"show": false (subtitle is hidden by default) "text": "Insert your subtitle here" "heading": "Heading2" "fontSize": 22 "bold": false (based on H2, font

		weight reduced i.e. bold disabled)
"outspacePane"	The filter pane	"backgroundColor": {"solid": {"color": "#FFFFFF"}} (\$ds_colour__background--primary) "titleSize": 12 "headerSize": 12 "searchTextSize": 12 These are smaller than the smallest font definition in the Design System (Small - 14px) due to the filter pane being a small element where font appears large "borderColor": {"solid": {"color": "#b3b3b3"}} (\$ds_colour__border) "checkboxAndApplyColor": {"solid": {"color": "#1a1a1a"}} (\$ds_colour__focus) This colour is applied when any applied filters include checkboxes, see example below

		 The screenshot shows the 'Filters' pane in Power BI. At the top is a search bar. Below it, 'Filters on this visual' is expanded. A 'Date' filter is applied with the range '26/05/2025 - 25/11/2...'. The 'Filter type' is set to 'Relative date'. The configuration shows items 'is in the last' 6 'months'. The 'Include today' checkbox is checked. An 'Apply filter' button is at the bottom right.
<code>"filterCard"</code>	Filter cards within the filter pane	<pre>"foregroundColor": {"solid": {"color": "#1a1a1a"}} "backgroundColor": {"solid": {"color": "#FFFFFF"}} "borderColor": {"solid": {"color": "#b3b3b3"}} "textSize": 12 "outspacePane"</pre> This is text colour on individual filter cards (\$ds_colour__text) (\$ds_colour__background--primary) (\$ds_colour__border) See note on text size above in

Buttons (`actionButton`), page navigators (`pageNavigator`) and bookmark navigator (`bookmarkNavigator`)

See: [Button - Design System](#)

The Design System defines 4 different button types, however Power BI only recognises 1 type of button with 4 states: Default, On hover, On press, and Disabled.

For this reason, there are limitations to the implementation of the custom theme on buttons. The Disabled state allows for implementation of the 'Disabled' button type, but for the rest, the 'Primary' button type styling has been applied.

To format buttons as 'Secondary' or 'As a link' types, manual styling is still required.

Unless specified, all button, page navigator and bookmark navigator text has the following attributes:

```
"verticalAlignment": "middle"
"horizontalAlignment": "center"
"fontFamily": "Roboto"
"fontSize": 14 (Small)
```

Unless specified, all button, page navigator and bookmark navigator fills have the following attributes:

```
"transparency": 0 (by default this is 50)
```

Button, page navigator and bookmark navigator state	Attributes
Default (" <code>\$id</code> ": " <code>default</code> ")	<u>Text</u> "fontColor": { "solid": { "color": "#ffffff" } } (\$ds_colour__background--primary) "bold": true <u>Button Fill</u> "fillColor": { "solid": { "color": "#0065bd" } } (\$ds_colour__link)
On hover (" <code>\$id</code> ":	<u>Text</u> "fontColor": { "solid": { "color":

"hover")	<pre>"#ffffff"}}</pre> (\$ds_colour__background--primary) <pre>"bold": true</pre> <u>Button Fill</u> <pre>"fillColor": {"solid": {"color":</pre> <pre>"#00437e"}}</pre> (\$ds_colour__link--hover)
On press ("\$id": "selected")	<u>Text</u> <pre>"fontColor": {"solid": {"color":</pre> <pre>"#1a1a1a"}}</pre> (\$ds_colour__text) <pre>"bold": true</pre> <u>Button Fill</u> <pre>"fillColor": {"solid": {"color":</pre> <pre>"#fdd522"}}</pre> (\$ds_colour__focus--background)
Disabled ("\$id": "disabled")	<u>Text</u> <pre>"fontColor": {"solid": {"color":</pre> <pre>"#1a1a1a"}}</pre> (\$ds_colour__text) <pre>"bold": false</pre> <u>Button Fill</u> <pre>"fillColor": {"solid": {"color":</pre> <pre>"#b3b3b3"}}</pre> (\$ds_colour__grey--medium)

Page navigators and bookmark navigators are formatted very similarly to buttons, though allow for extra customisation that buttons do not have. One example is the "accentBar" , which the Design System contains on the bottom of 'selected' buttons. It is not possible to add this to buttons in Power BI, but this feature has been included in page and bookmark navigators.

Page navigator and bookmark navigator state	Attributes
On press ("\$id": "selected")	<u>Accent Bar</u> <pre>"show": true</pre> <pre>"color": {"solid": {"color": "#000000"}}</pre>

	<code>"position": "Bottom"</code> <code>"width": 5</code>
--	--

Tables (`tableEx`)

Unless specified, text has been formatted as:

```
"fontFamily": "Roboto"  
"fontSize": 16  
"color": "#1a1a1a"
```

Table element	
Column header background colour	<code>"backColor": {"solid": {"color": "#ebebeb"}}</code> (\$ds_colour__background--tertiary)
Primary row background colour	<code>"backColorPrimary": {"solid": {"color": "#ffffff"}}</code> (\$ds_colour__background--primary)
Secondary row background colour	<code>"backColorSecondary": {"solid": {"color": "#f8f8f8"}}</code> (\$ds_colour__background--secondary)
Column header text	<code>"bold": true</code>
Other text	<code>"bold": false</code>
Table outline colour	<code>"outlineColor": {"solid": {"color": "#b3b3b3"}}</code> (\$ds_colour__border)